

ICT2217

Network Security

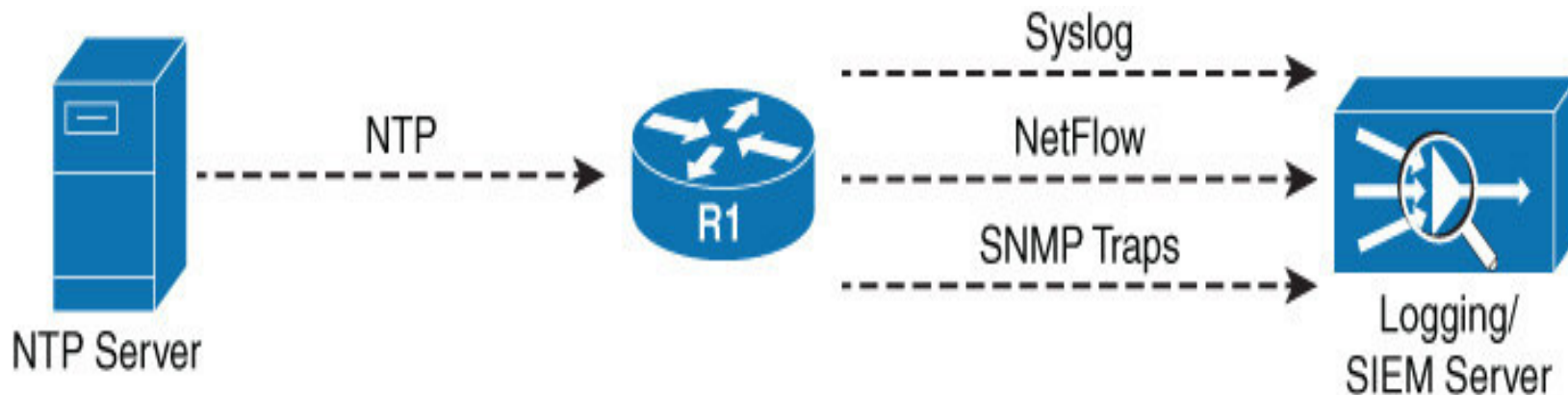
Lab 3.5 Notes:

Network Security Monitoring using Syslog with NTP



Finally, with the various network security implemented, we are now ready to perform **network security monitoring** by **logging** all alerts to centralised servers for analysis and reporting.

Logging is commonly implemented using **Syslog**, **SNMP** and/or **NetFlow** protocols to centralised **servers** or **SIEM** (Security Information and Event Management) systems.



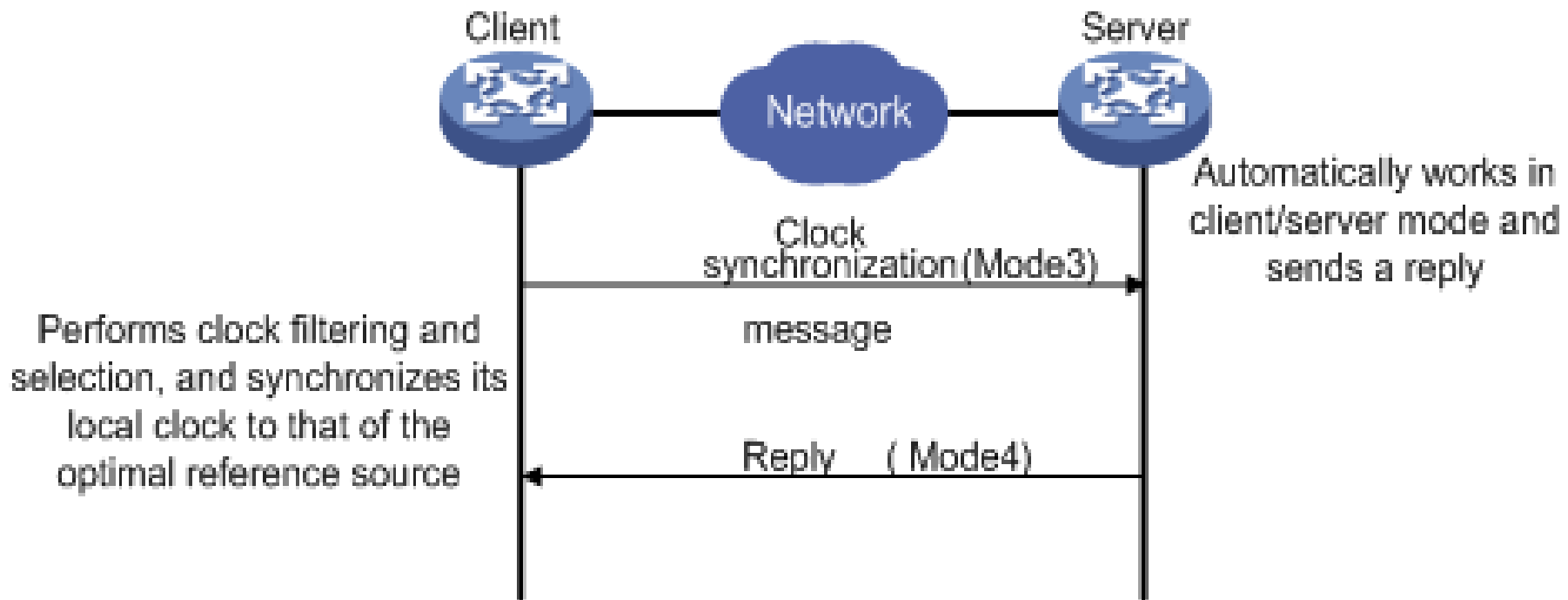
To facilitate analysis of log data, it is essential to **synchronize time** between all network devices, which can be achieved by using **NTP** (Network Time Protocol).

1

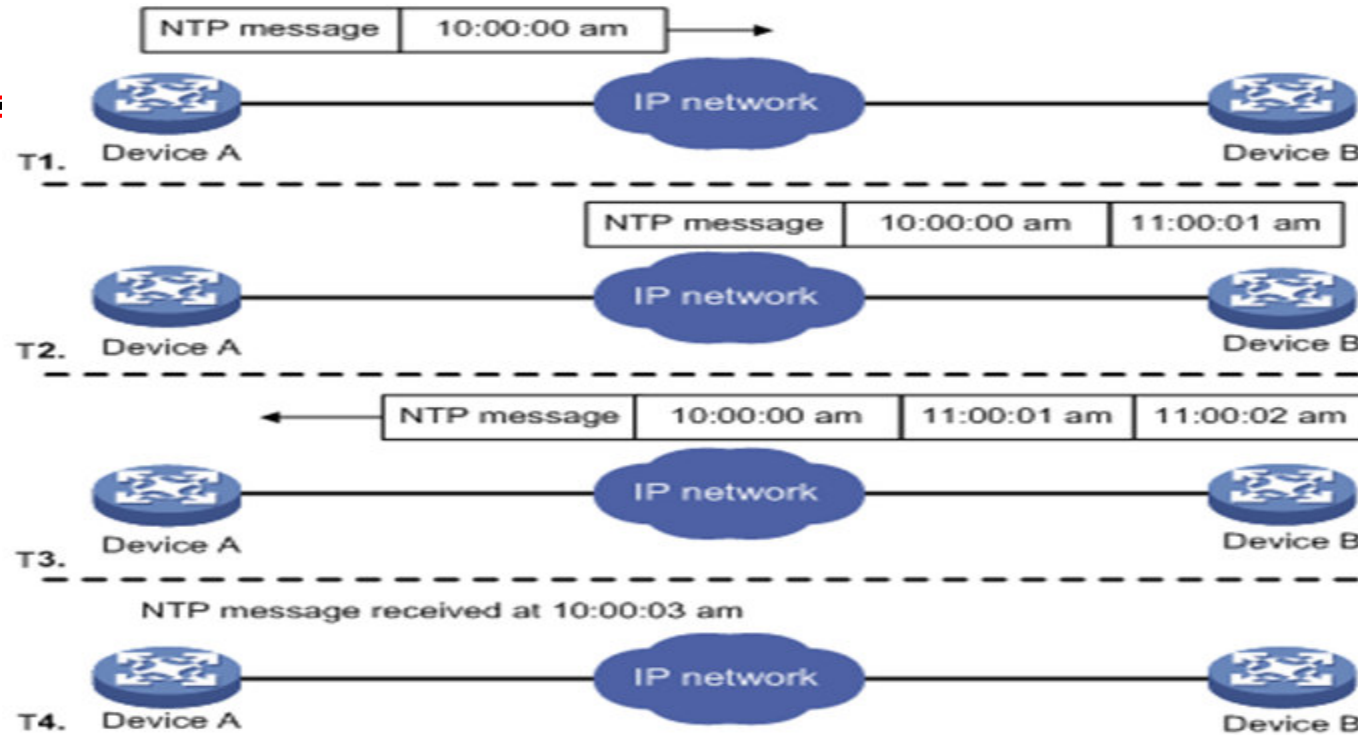
Configure **NTP** for synchronizing time between network devices to facilitate analysis of log data.

As defined in RFC 5905, there are 3 **NTP** operation modes – **client/server**, peer and broadcast. We'll discuss the common client/server mode to understand how NTP works.

NTP operates over **UDP port 123**. In **client/server mode**, the NTP client initiates a request to the NTP server, and receives a reply as shown:



In **client/server mode**, NTP client synchronises its clock by performing the following computation based on NTP reply:

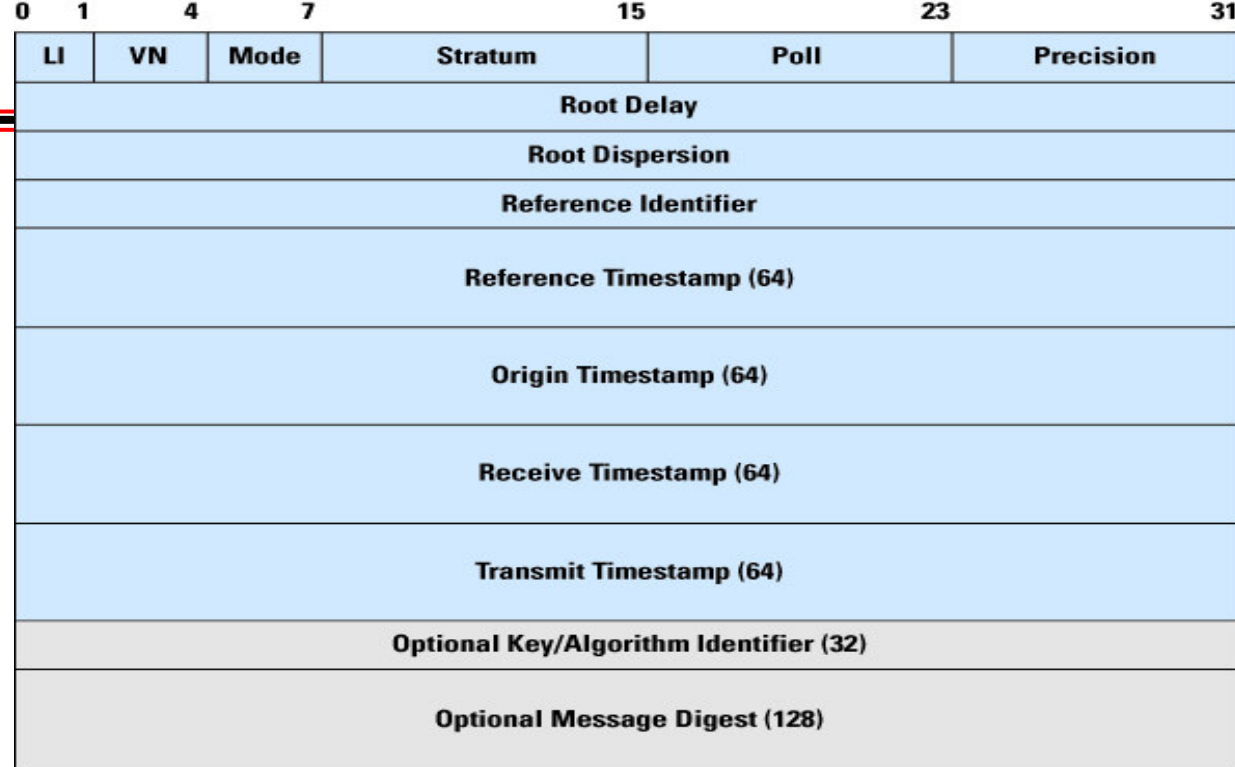


Offset of client clock from server clock: $\theta = \frac{(T2 - T1) + (T3 - T4)}{2}$ e.g. $\theta = \frac{1h1s + 59m59s}{2} = 1h$

Round-trip delay: $\delta = (T4 - T1) - (T3 - T2)$ e.g. $\delta = 3s - 1s = 2s$

(Clock filter and mitigation algorithms will then be used to determine the best and final offset to synchronize the system clock.)

To support NTP computation, NTP client and server register its timestamps in the **NTP message format** as follows:



	<u>NTP request</u>	<u>NTP reply</u>
Origin timestamp:	ignore	T1
Receive timestamp:	ignore	T2
Transmit timestamp:	T1	T3

Reference timestamp: time when system clock was last updated

An example of NTP request sent by NTP client:

The image shows a Wireshark packet capture window titled 'ntp-key1.pcapng'. The packet list pane shows two packets: packet 8.649520 (NTP Version 4, client) and packet 8.650151 (NTP Version 4, server). The packet details pane for the selected packet (8.649520) shows the following structure:

- Flags: 0xe3, Leap Indicator: unknown (clock unsynchronized), Version number: NTP Version 4, Mode: client
 - 11.. = Leap Indicator: unknown (clock unsynchronized) (3)
 - ..10 0... = Version number: NTP Version 4 (4)
 -011 = Mode: client (3)
- Peer Clock Stratum: unspecified or invalid (0)
- Peer Polling Interval: 6 (64 sec)
- Peer Clock Precision: 0.000977 sec
- Root Delay: 0 seconds
- Root Dispersion: 0.0009613037109375 seconds
- Reference ID: (Initialization)
- Reference Timestamp: Nov 9, 2018 07:54:19.573000093 UTC
- Origin Timestamp: Nov 9, 2018 08:14:51.533362830 UTC
- Receive Timestamp: Nov 9, 2018 08:14:51.574000094 UTC
- Transmit Timestamp: Nov 9, 2018 08:15:55.569000093 UTC
- Key ID: 00000001
- Message Authentication Code: 81051e78be04d9d904382e6c52d1a706

The packet bytes pane shows the raw data in hexadecimal and ASCII. The first three bytes are 01 01 00, which correspond to the NTP flags 0xe3 (11100111 in binary). The ASCII column shows the corresponding characters: '{', '{', and 'L'.

Mode (ntp.flags.mode), 1 byte | Packets: 134 · Displayed: 134 (100.0%) | Profile: Default

An example of NTP reply sent by NTP server:

The image shows a Wireshark packet capture window titled 'ntp-key1.pcapng'. The packet list shows two NTP packets: a client request (No. 8.649520) and a server reply (No. 8.650151). The server reply packet is selected, and its details are shown in the packet details pane. The details pane shows the NTP Version 4 structure, including the Leap Indicator, Version number, Mode, Peer Clock Stratum, Peer Polling Interval, Peer Clock Precision, Root Delay, Root Dispersion, Reference ID, Reference Timestamp, Origin Timestamp, Receive Timestamp, Transmit Timestamp, Key ID, and Message Authentication Code. The packet bytes pane shows the raw data of the packet, with the first 12 bytes highlighted in blue.

No.	Time	Source	Destination	Protocol	Length	Info
8.649520		192.168.1.2	192.168.1.1	NTP	110	NTP Version 4, client
8.650151		192.168.1.1	192.168.1.2	NTP	110	NTP Version 4, server

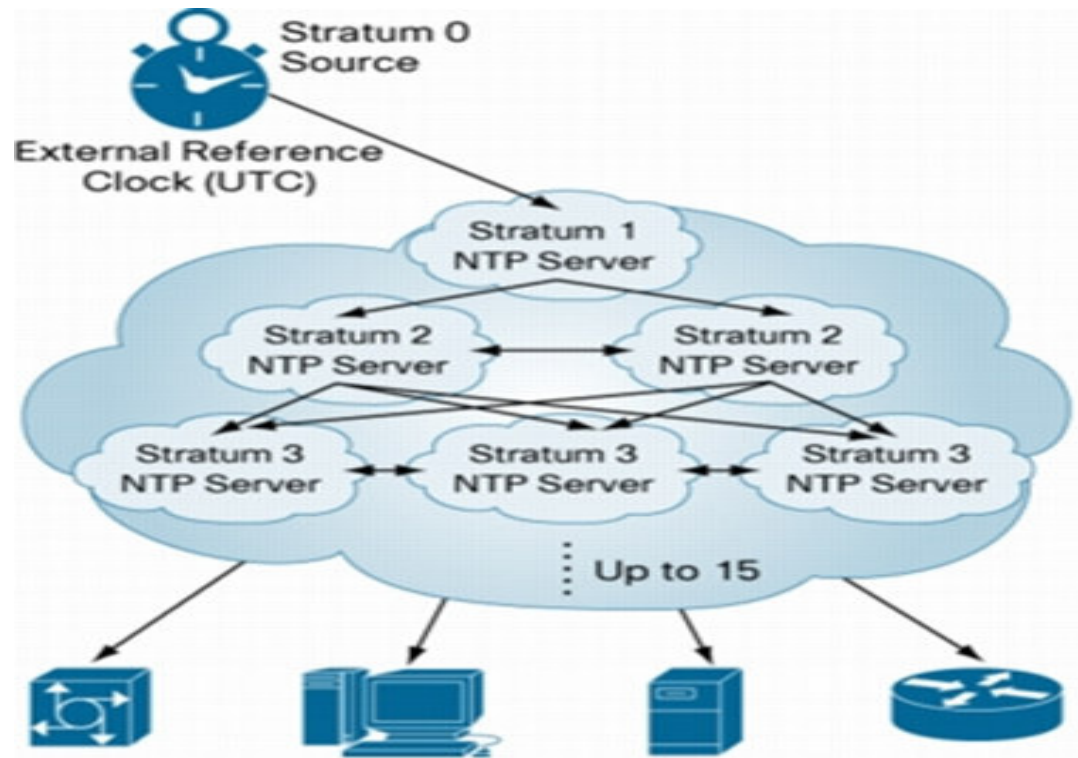
Flags: 0x24, Leap Indicator: no warning, Version number: NTP Version 4, Mode: server
00.. = Leap Indicator: no warning (0)
..10 0... = Version number: NTP Version 4 (4)
.... .100 = Mode: server (4)
Peer Clock Stratum: secondary reference (3)
Peer Polling Interval: 6 (64 sec)
Peer Clock Precision: 0.000000 sec
Root Delay: 0.0522918701171875 seconds
Root Dispersion: 7.96548461914063 seconds
Reference ID: 183.177.72.201
Reference Timestamp: Nov 9, 2018 08:15:28.537370376 UTC
Origin Timestamp: Nov 9, 2018 08:15:55.569000093 UTC
Receive Timestamp: Nov 9, 2018 08:15:55.529970338 UTC
Transmit Timestamp: Nov 9, 2018 08:15:55.530000587 UTC
Key ID: 00000001
Message Authentication Code: 862dbd4bcf2f9c370aa08c50de6b9d8c

0020 01 02 00 7b 00 7b 00 4c 84 14 24 03 06 eb 00 00 ...{·{·L··\$·.....
0030 0d 63 00 07 f7 2a b7 b1 48 c9 df 8f c0 a0 89 91 ·c··*··H·.....
0040 1a df df 8f c0 bb 91 a9 fd 78 df 8f c0 bb 87 ac ·······x·.....

Mode (ntp.flags.mode), 1 byte | Packets: 134 · Displayed: 134 (100.0%) | Profile: Default

To be scalable, **NTP servers** are typically setup in an hierarchical structure to support time synchronisation in practice.

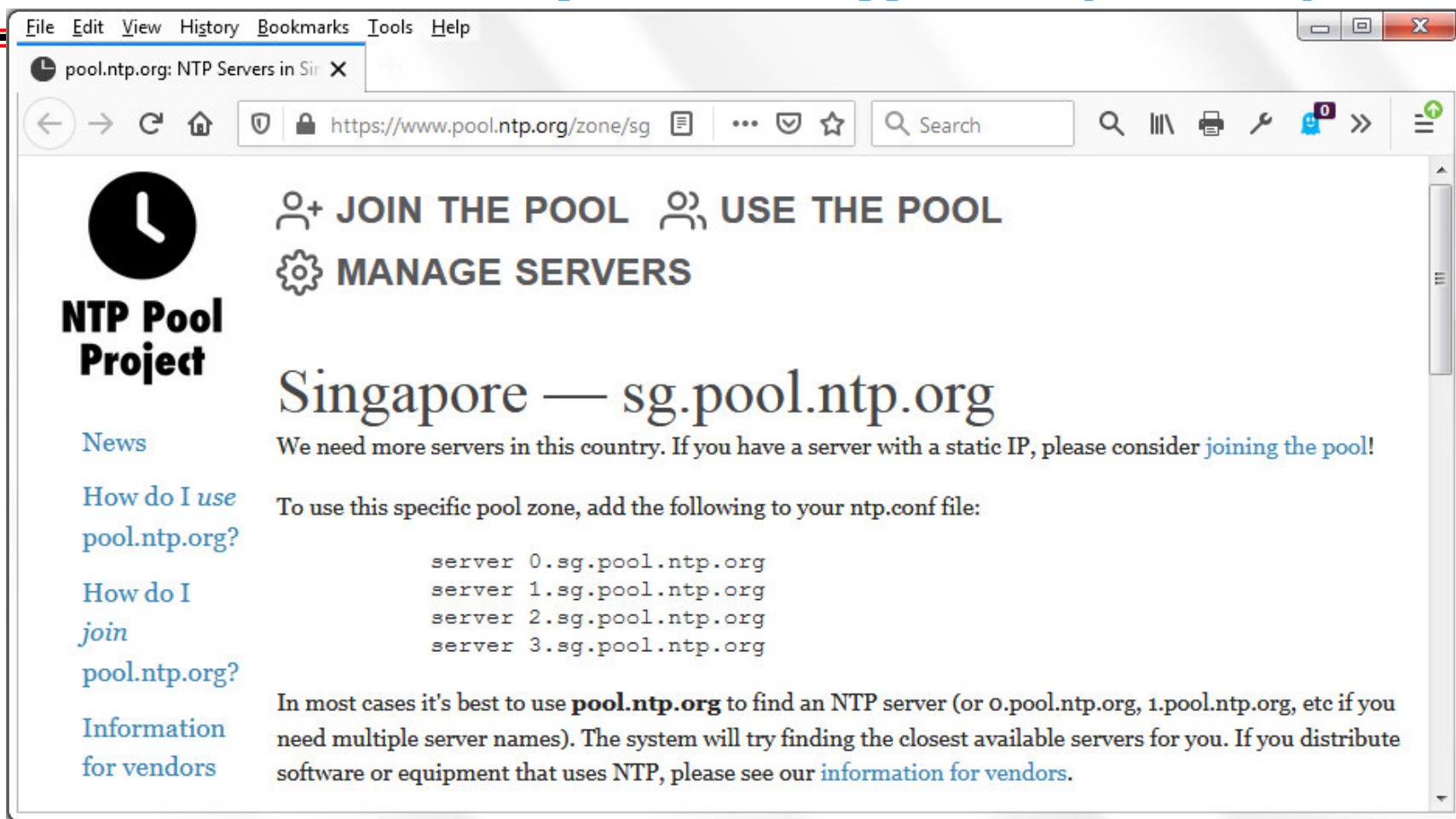
NTP uses the concept of **stratum** to describe how many **NTP hops** away a machine is from the most reliable authoritative time source at stratum 0.



As the stratum number increases, its accuracy degrades.

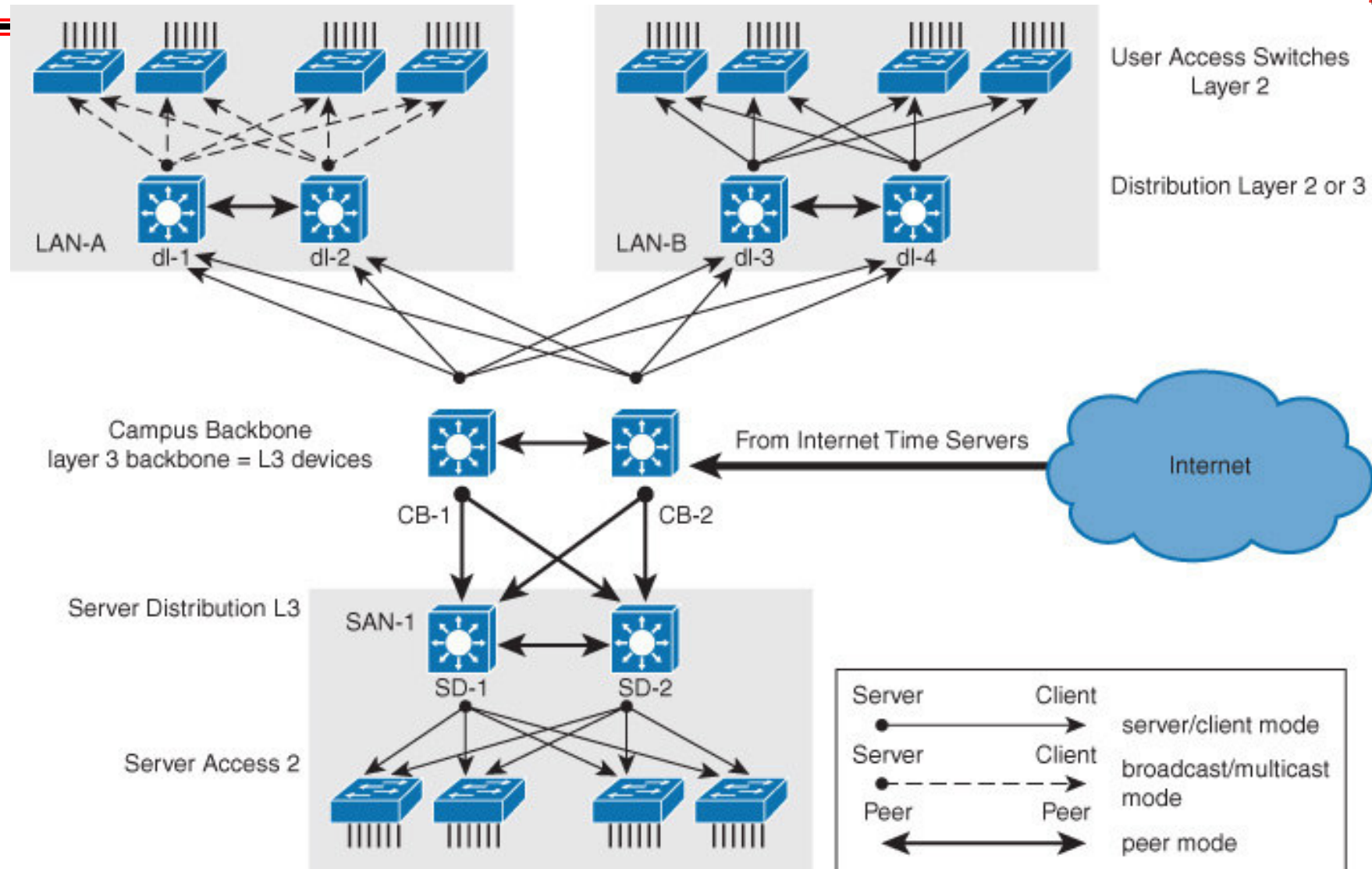
UTC (Coordinated Universal Time), aka GMT (Greenwich Mean Time), is the current worldwide standard for time and date.

An example is the [NTP pool project](https://www.ntppool.org/zone/sg) which is maintaining a network of [NTP servers](https://www.ntppool.org/zone/sg) that are being used by many systems around the world. <https://www.ntppool.org/zone/sg>



To use the pool, simply configure your device to point to any of the provided domain name, each representing a random set of NTP servers. 10

After obtaining time from an NTP pool, an **enterprise** may implement time synchronization within its network in an **hierarchical** manner:



To be specific, we'll now discuss how to configure the network devices to synchronize time using **NTP**.

Before configuring NTP, you may wish to configure your device **clock** to current date/time:

```
R1# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)# clock timezone SGT 8
R1(config)# ^Z
R1#
R1# clock set 20:52:00 25 October 2017
*Oct 25 12:52:00.000: %SYS-6-CLOCKUPDATE: System clock has been updated from 20:36:38
SGT Wed Oct 25 2015 to 20:52:00 SGT Wed Oct 25 2015, configured from console by
console.
R1# show clock
20:52:55.051 SGT Wed Oct 25 2017
```

Any letters representing your
timezone, e.g. SGT for Singapore
Time, which is UTC + 8 hrs.

Based on the design of your network, configure network devices as **NTP server**, **client**, **peer** or **broadcast**:



- **Server**: provides time information to clients, only used if you are not able to reach reliable external NTP servers

```
R1 (config) # ntp master [stratum]
```

- **Client**: synchronizes time with an NTP server

```
R1 (config) # ntp server {ip | hostname} [version] [prefer]
```

- **Peer**: also called symmetric mode, which work both as server and client to synchronize, or be synchronized by each other

```
R1 (config) # ntp peer {ip | hostname} [version] [prefer]
```

- **Broadcast**: pushes time announcements from NTP server to clients, used only when time accuracy is not a concern

Server to broadcast:

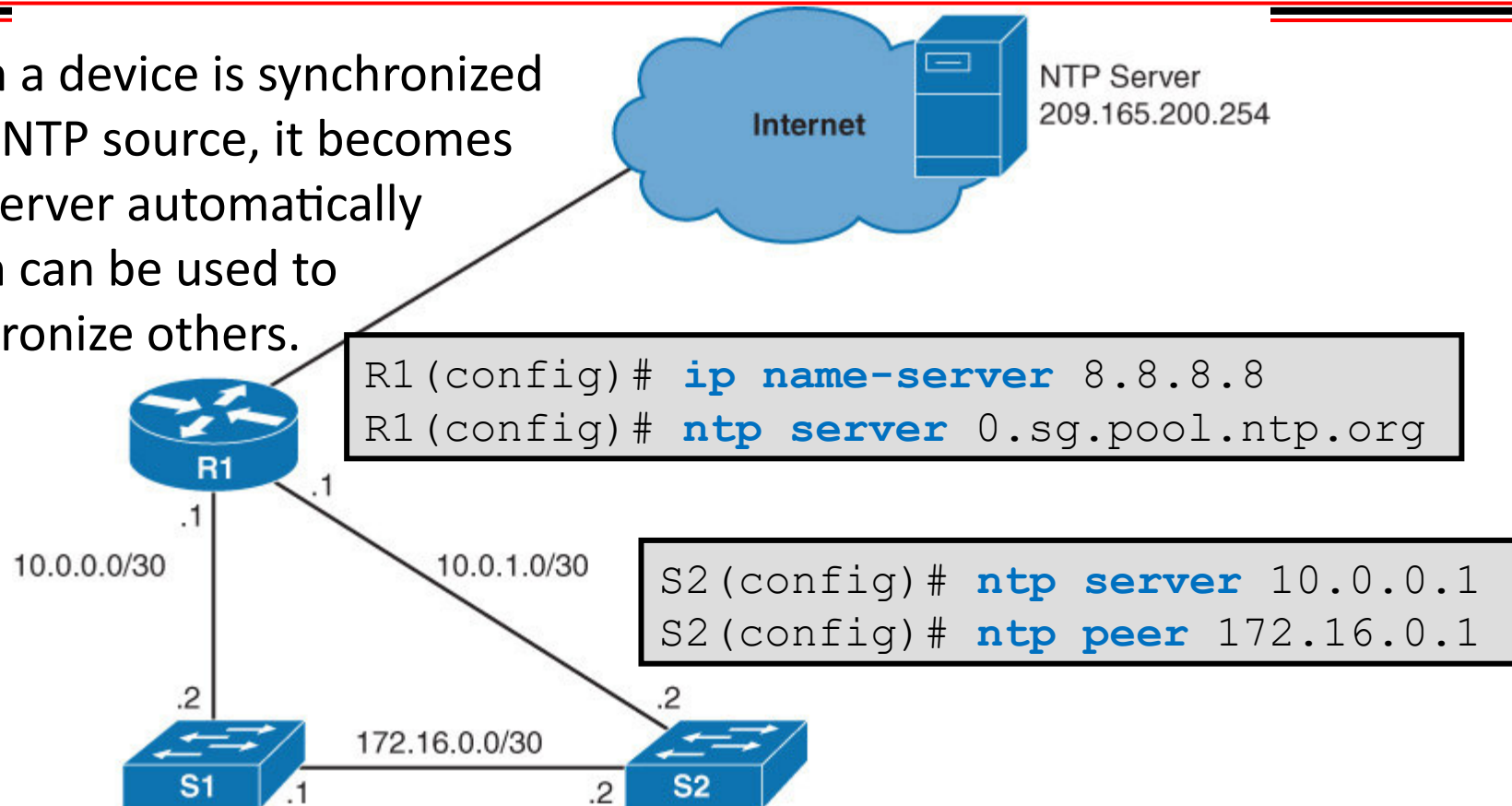
```
R1 (config-if) # ntp broadcast [version]
```

Client to receive:

```
R1 (config-if) # ntp broadcast client
```


An example of configuring **NTP**, with edge router R1 synchronizing with an external NTP time source, which in turn acts as NTP server for internal switches.

When a device is synchronized to an NTP source, it becomes NTP server automatically which can be used to synchronize others.



```
R1(config)# ip name-server 8.8.8.8
R1(config)# ntp server 0.sg.pool.ntp.org
```

```
S2(config)# ntp server 10.0.0.1
S2(config)# ntp peer 172.16.0.1
```

```
S1(config)# ntp server 10.0.0.1
S1(config)# ntp peer 172.16.0.2
```


To **verify** that the device's clock has indeed been synchronized with NTP, use the following show commands:

To verify which NTP server your device is synchronizing with, e.g.:

```
R1# show ntp associations
```

address	ref clock	st	when	poll	reach	delay	offset	disp
*~209.165.200.254	.LOCL.	1	24	64	17	1.000	-0.500	2.820

* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

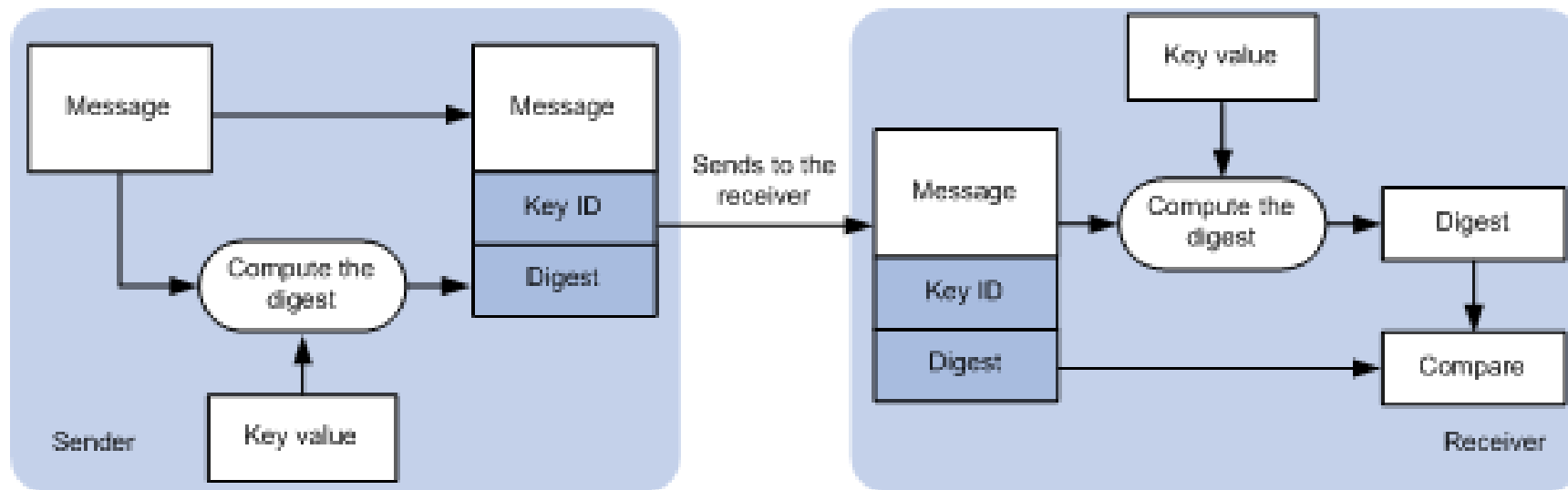
By synchronizing with stratum 1 time source, your device automatically becomes stratum 2 time source as shown, e.g.:

```
R1# show ntp status
```

```
Clock is synchronized, stratum 2, reference is 209.165.200.254  
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**10  
ntp uptime is 1500 (1/100 of seconds), resolution is 4000  
reference time is D67E670B.0B020C68 (05:22:19.043 PST Mon Jan 13 2014)  
clock offset is 0.0000 msec, root delay is 0.00 msec  
root dispersion is 630.22 msec, peer dispersion is 189.47 msec  
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000000 s/s  
system poll interval is 64, last update was 5 sec ago.
```

Unfortunately, an attacker can **inject false NTP replies** to confuse time synchronization and log timings. Thus, it is advisable to enable **NTP authentication** to protect NTP clients.

Both NTP client and server can be configured with the same **pre-shared key/password** which is used to compute **MD5 hash** for **authentication**:



Using the **pre-shared key**, **NTP server** generates an MD5 hash and send it together with NTP message.

Using the same **pre-shared key**, **NTP client** re-computes the MD5 hash and verifies it with the received MD5 hash before accepting the NTP message.

For configuring of **NTP authentication**, the additional commands required are as follows:

- Define one or more NTP key(s), each number specifies a unique NTP key (password):

```
R1 (config) # ntp authentication-key key-number md5 string
```

- Specify which key to enable for authentication:

```
R1 (config) # ntp trusted-key key-number
```

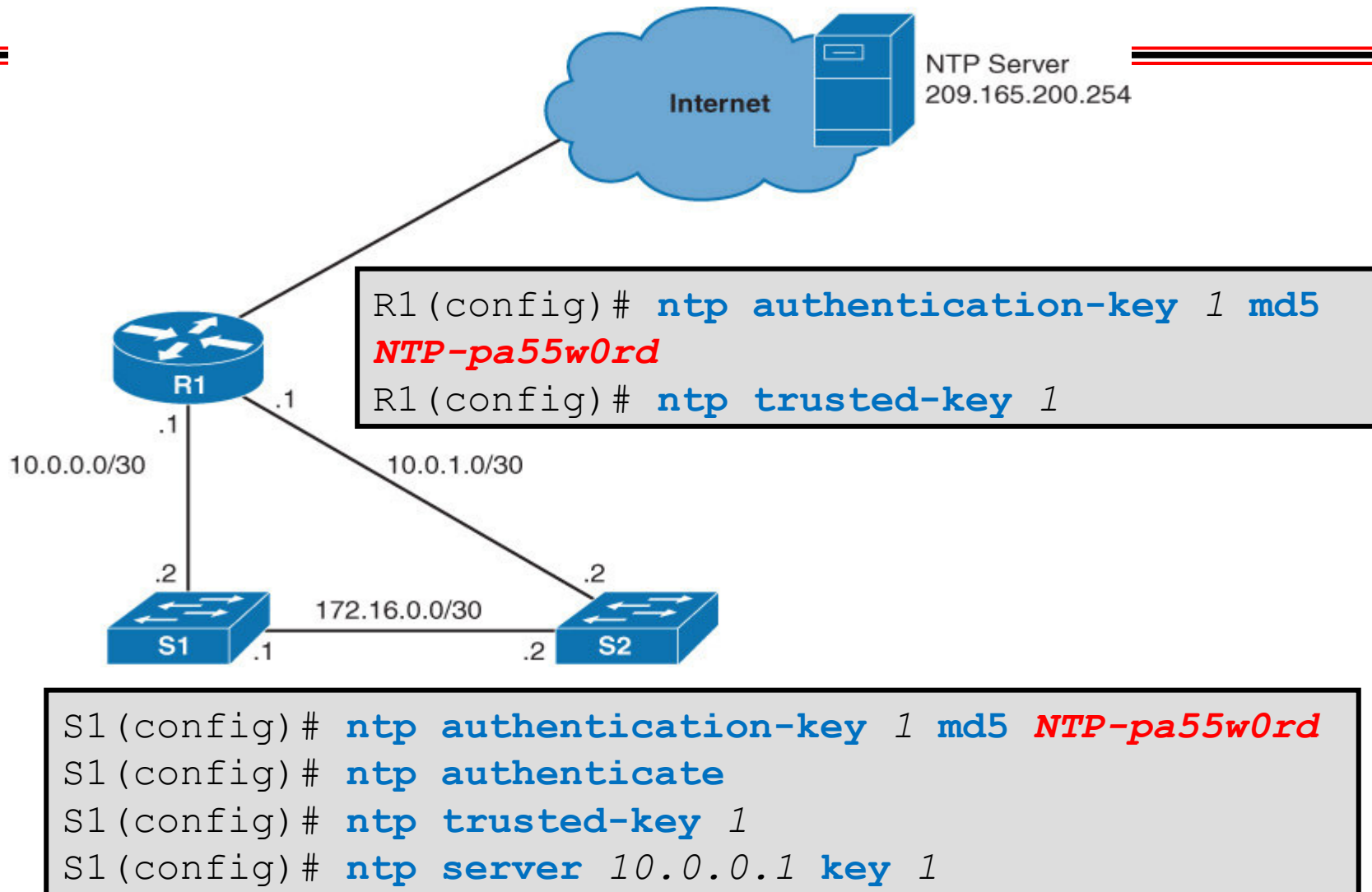
- Enable NTP authentication:

```
R1 (config) # ntp authenticate
```

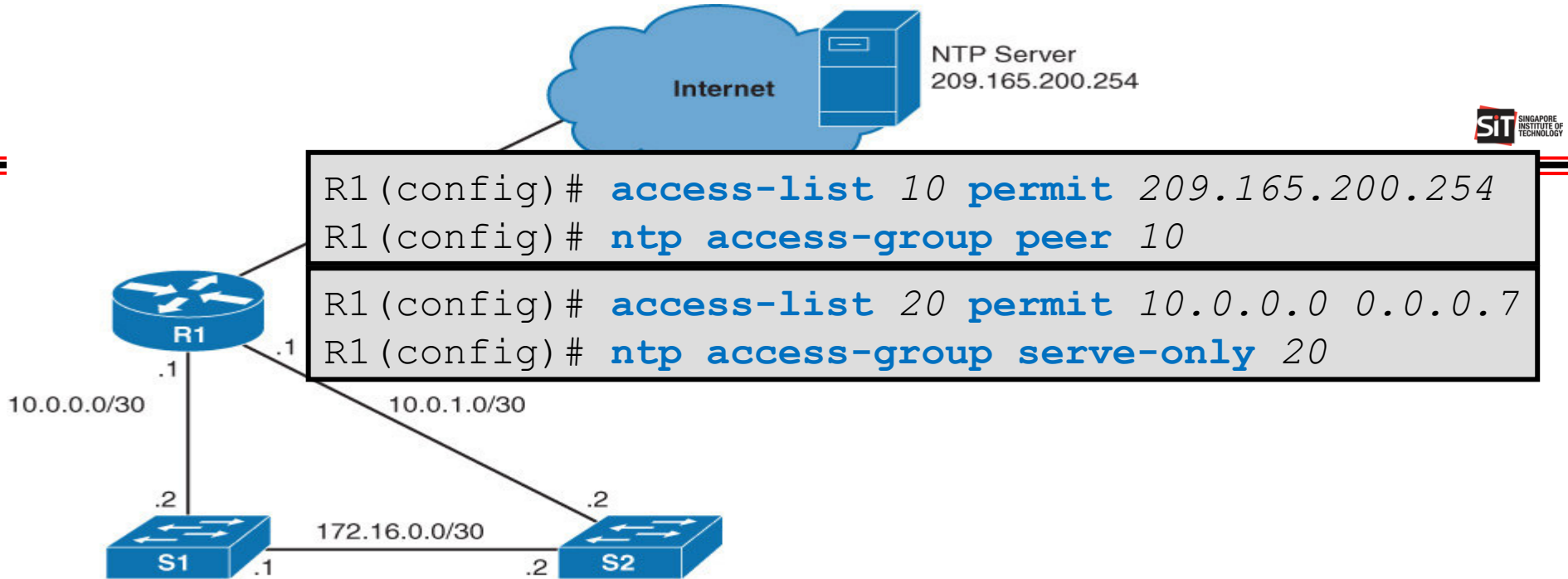
- For NTP client to specify which key to use for authentication:

```
R1 (config) # ntp server {ip | hostname} key key-number
```

An example of configuring **NTP authentication** to prevent NTP clients from synchronising with attacker's time source.



In addition, NTP can be protected by ACLs.



Command for configuring ACL:

```
ntp access-group {query-only | serve-only | serve | peer}
access-list-number
```

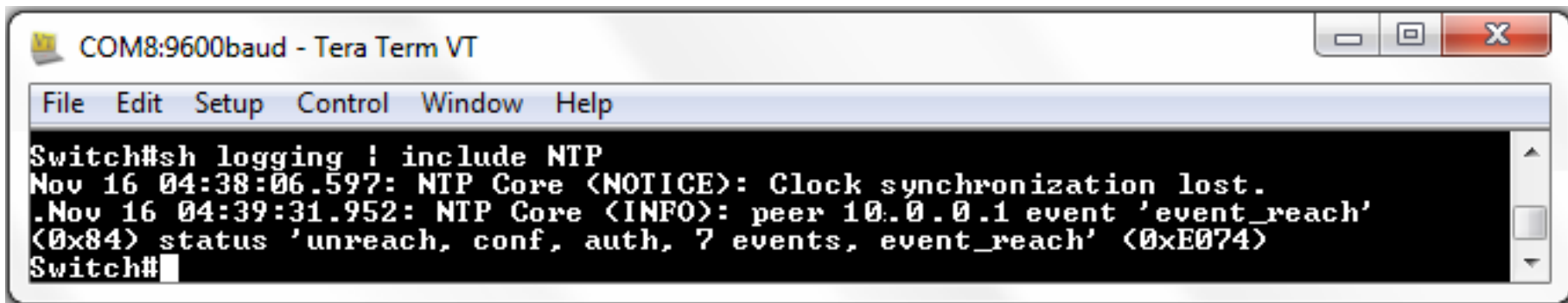
- **peer**: accept NTP reply and also respond to NTP request
- **serve-only**: for server to respond to NTP request only
- **serve**: similar as 'serve-only', but also accept NTP control queries
- **query-only**: only accept NTP control queries such as SNMP

Since NTP is important, it is also recommended to enable **NTP logging** to log significant NTP events.

Enabling NTP logging (logging centrally will be discussed shortly):

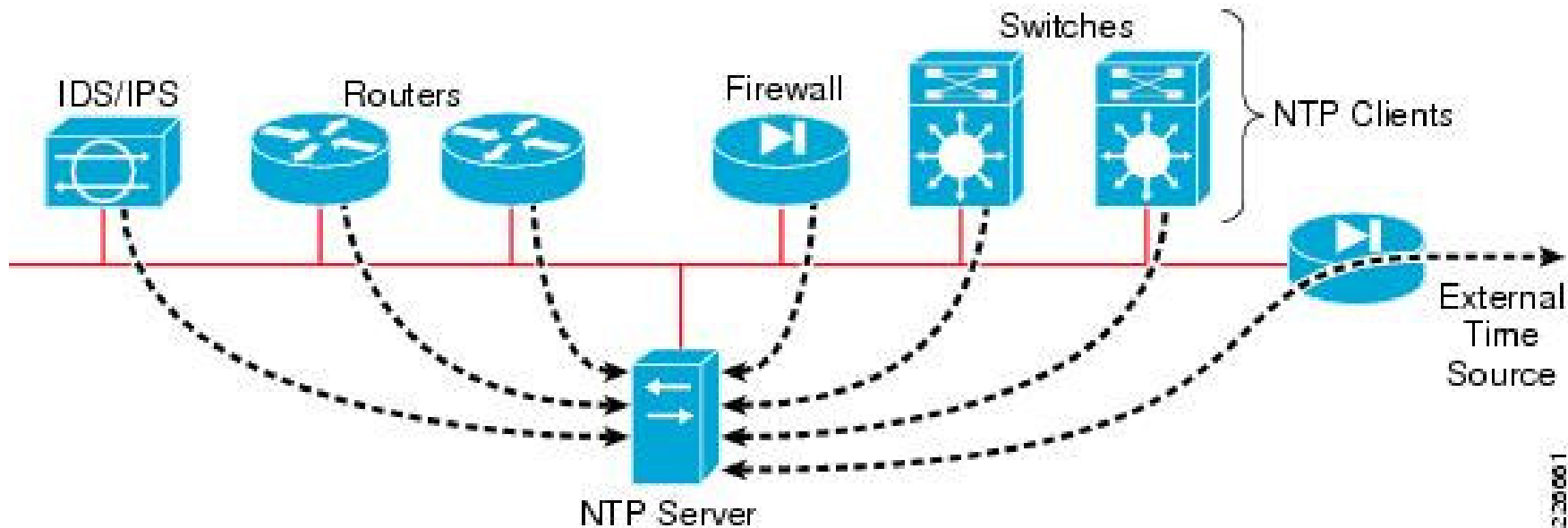
```
Switch(config) # ntp logging
```

Examples of NTP logs:



```
COM8:9600baud - Tera Term VT
File Edit Setup Control Window Help
Switch#sh logging ! include NTP
Nov 16 04:38:06.597: NTP Core <NOTICE>: Clock synchronization lost.
Nov 16 04:39:31.952: NTP Core <INFO>: peer 10.0.0.1 event 'event_reach'
<0x84> status 'unreach, conf, auth, 7 events, event_reach' <0xE074>
Switch#
```


If **out-of-band (OOB) management network** is implemented, it is more direct and secure to perform **NTP time synchronization** over OOB, e.g.:



If **VRF** is used: (note that commands are different for different devices)

```
R1/S1(config)# ntp server vrf Mgmt-vrf 10.0.0.1
```

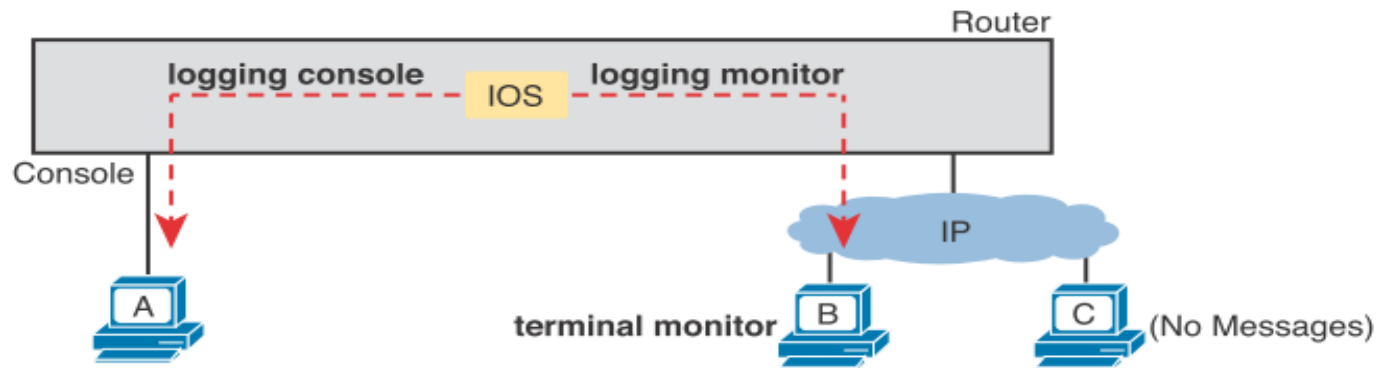
```
FW(config)# ntp server 10.0.0.1 source management
```

- 2 Configure **Syslog** which is one of the most commonly implemented method for logging.

Syslog (System message logging) (RFC 5424) is a standard protocol for event notification messages which is widely adopted by the industry.

While configuring Cisco devices in the labs, you must have seen or even be annoyed by the **syslog messages** that appeared, e.g.

```
*Dec 18 17:10:15.079: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down
```



Disable syslog on console:

```
R1 (config) # no logging console
```

Prevent syslog from interrupting command entry:

```
R1 (config) # logging synchronous
```

Enable syslog over Telnet/SSH:

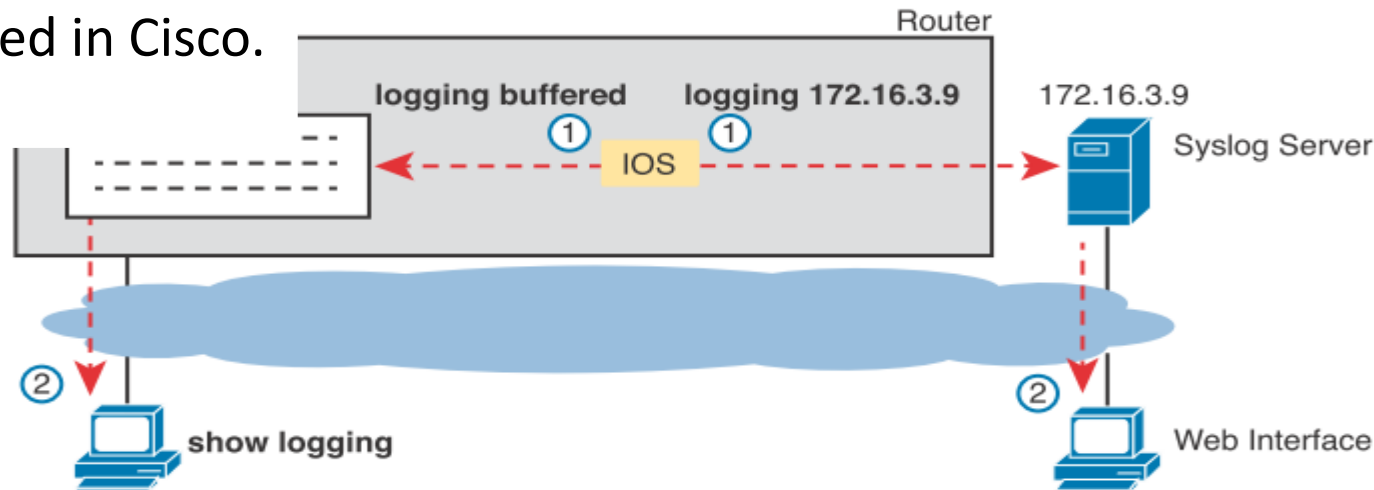
```
R1 (config) # logging monitor
```

In addition, at B:

```
R1 # terminal monitor
```

For monitoring of network security, it is more useful to store log messages in the device RAM, or even better to a **centralised Syslog server**.

By default, **Syslog** messages are transported over **UDP port 514** (RFC 5426) or more securely over **TLS over TCP port 6514** (RFC 5425) – may not be supported in Cisco.



To log messages into RAM:

```
R1 (config) # logging buffered
```

In addition:

```
R1# show logging
```

To log messages to Syslog server:

```
R1 (config) # logging  
172.16.3.9  
R1 (config) # logging trap 4
```

All errors 4 or below (more serious) are logged.

To prevent cluttering the log with too much data, **Syslog** has a **severity level** rating which you can use it to control which messages to log.

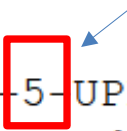
Keyword	Numerical	Description	
Emergency	0	System unusable	Severe
Alert	1	Immediate action required	
Critical	2	Critical Event (Highest of 3)	Impactful
Error	3	Error Event (Middle of 3)	
Warning	4	Warning Event (Lowest of 3)	
Notification	5	Normal, More Important	Normal
Informational	6	Normal, Less Important	
Debug	7	Requested by User Debug	Debug

Service	To Set Message Levels
Console	logging console <i>level-name</i> <i>level-number</i>
Monitor	logging monitor <i>level-name</i> <i>level-number</i>
Buffered	logging buffered <i>level-name</i> <i>level-number</i>
Syslog	logging trap <i>level-name</i> <i>level-number</i>

An example of **Syslog message** which shows the date/time of occurrence, the severity level and the description.

*Dec 18 17:10:15.079: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down

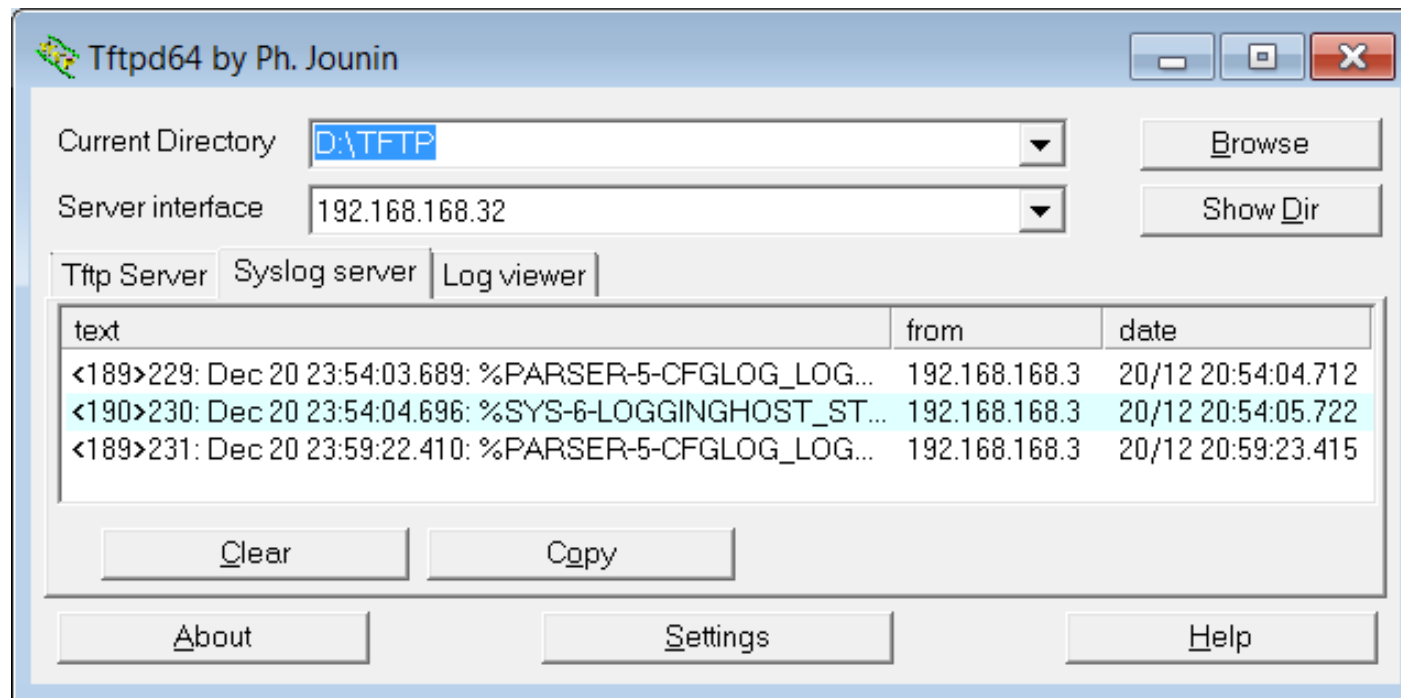
severity level



- **A timestamp:** *Dec 18 17:10:15.079
- **The facility on the router that generated the message:** %LINEPROTO
- **The severity level:** 5
- **A mnemonic for the message:** UPDOWN
- **The description of the message:** Line protocol on Interface FastEthernet0/0, changed state to down

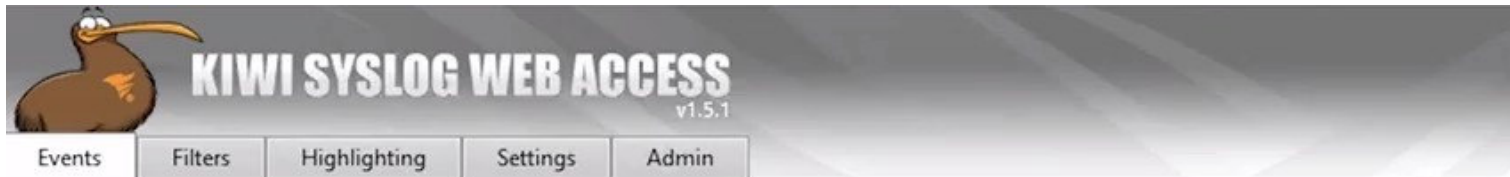
There are many [syslog servers](#) available. An example is the free open source tftpd64 server which can also function as syslog server that you can try in the lab.

<https://pjo2.github.io/tftpd64/>



Alternatively, you may also try other Syslog servers like Kiwi Syslog which offers a free edition supporting up to 5 clients.

<https://www.solarwinds.com/free-tools/kiwi-free-syslog-server>



The screenshot shows the Kiwi Syslog Web Access v1.5.1 interface. At the top left is a cartoon kiwi bird logo. The title 'KIWI SYSLOG WEB ACCESS v1.5.1' is prominently displayed. Below the title are five tabs: 'Events', 'Filters', 'Highlighting', 'Settings', and 'Admin'. The 'Events' tab is selected. Below the tabs is a filter dropdown menu showing '[All Kiwi Syslog Server Events]'. The main area contains a table of log events with columns for Date, Time, Facility, Level, Host Name, and Message Text.

Date	Time	Facility	Level	Host Name	Message Text
2015-03-20	14:05:00	Cron	Warning	10.111.201.39	Logon failed
2015-03-20	14:05:00	Local1	Warning	10.111.201.39	Logoff Successful
2015-03-20	14:05:00	Local7	Critical	10.111.201.39	This is a test message generated by Kiwi SyslogGen
2015-03-20	14:05:00	Local1	Alert	10.111.201.39	Logon successful
2015-03-20	14:04:59	Cron	Error	10.111.201.39	Logon failed
2015-03-20	14:04:59	System1	Critical	10.111.201.39	Logoff Successful
2015-03-20	14:04:59	Syslog	Error	10.111.201.39	This is a test message
2015-03-20	14:04:59	System0	Critical	10.111.201.39	Logon successful
2015-03-20	14:04:58	Cron	Warning	10.111.201.39	Logon failed
2015-03-20	14:04:58	System2	Info	10.111.201.39	Logoff Successful
2015-03-20	14:04:58	User	Notice	10.111.201.39	This is a test message
2015-03-20	14:04:58	News	Debug	10.111.201.39	Logon successful
2015-03-20	14:04:57	Cron	Info	10.111.201.39	Logon failed

